Circular Economy Dynamics in the Apparel Industry: The Emergence of "Retrovation"

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1 Introduction

The apparel industry, renowned for the heavy environmental impact of the production process, should be at the forefront of a significant transformation, driven by the urgency to meet environmental and social sustainability goals (Pal et al., 2019; Abbate et al., 2023). In order to reinvent the apparel value chain, a variety of actions is expected, not only modifying the upstream stages of the production process, as well as the design, the supplier relations, the product's end of life but also raising the sensitivities of the various stakeholders involved, not underestimating the role played by an increased customers awareness of the side effects of overconsumption, and by the pressures of social movements in rethinking the overall apparel ecosystem toward sustainability (Sedita et al., 2024). The implementation of circular economy practices might be considered one possible solution to accelerate this (Bassett, 2021; Jia et al., 2020). Circular economy, in contrast to the traditional linear economy, entails the adoption of sustainable business models (SBMs) (Bocken et al., 2014) where economic development is separated from resource consumption by reducing the use and recirculating natural resources (Corona et al., 2019). It encompasses designing out waste, maintaining products and materials in use, and regenerating natural systems. However, it is important to recognize that not all firms in the apparel industry are currently implementing circular practices. Despite the widespread awareness of the need for circularity, the adoption of such practices remains uneven, with some companies still adhering to traditional linear models of production.

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The circular economy is grounded in the following four fundamental principles derived from nature (Ellen MacArthur Foundation, 2012; Weetman, 2016; Gardetti, 2019): (1) Waste as nutrients: like in nature, waste becomes food or nutrients. Objects are reused and redesigned to extend their life cycle indefinitely. (2) Resilience: built through diversity, the circular economy uses a wide pool of resources to adapt to different contexts and circumstances. (3) Renewable energy: a system powered by renewable energy ensures a continuous flow of resources, ideas, and information. (4) Systemic approach: interconnected elements create opportunities and set examples for other economies, cultures, and societies.

Building on such principles, we suggest that a notable trend within the shift toward circularity is the resurgence of traditional practices and processes in innovative contexts, a phenomenon which has been called "retrovation" (Suominen & Sivula, 2016). This is particularly relevant in our case because, in addition to innovation, retrovation—by its very nature—frequently involves reviving and modernizing heritage techniques and craftsmanship in ways that can also align with contemporary sustainability objectives (Sedita, 2024). This study explores the complex interplay between circularity and retrovation, aiming to illustrate how this approach can foster sustainable transformations in the apparel industry.

Particularly, by examining the convergence of craftsmanship, traditional production methods, and modern sustainability orientation of firms in the apparel sector, we seek to highlight the potential of heritage practices to drive circular sustainability through innovation. With empirical analysis, we focus on an exemplary case of circular sustainability where we also suggest retrovation is successfully implemented. This case serves as beacons of change, showcasing how integrating traditional knowledge with modern environmental and pro-social goals can lead to significant advancements in sustainability. In fact, our focus on this case is intended to illustrate the broader potential of retrovation to drive change within the apparel sector. While not all firms are currently adopting circular practices, the success story we examine can provide a roadmap for others in the industry to follow.

The focus on this sector is important because it has experienced unprecedented growth, producing between 100 and 150 billion items of clothing annually (McKinsey & Company, 2022; World Economic Forum, 2021). Several businesses in this sector have sped up the season cycles; they manufacture their clothes in other countries, mainly Eastern ones where production costs and labor rights are lower, and they sometimes adopt low levels of transparency in their communications (Gardetti, 2019). This surge in production has doubled the application of textiles since the turn of the millennium, yet garment utilization has drastically decreased, leading to significant waste (MacArthur, 2017). In 2018, the fashion industry was responsible for 2.1 billion tons of CO_2 emissions, roughly 4% of the global total, with some estimates placing this impact as high as 8–10% (Berg et al., 2020). Upstream activities, such as raw material production and processing, contribute around 70% of this impact, while downstream activities like product use account for the remaining 30% (Global Fashion Agenda and McKinsey & Company, 2020).

In response to these environmental and social challenges, some companies in the (global) apparel industry are increasingly adopting eco-fashion practices and

sustainable marketing strategies (Joergens, 2006; Fletcher, 2008; Chan & Wong, 2012). This shift is not limited to slow fashion and luxury brands but extends to a part of the fast fashion companies as well, which are known for their cost-efficiency and rapid market turnaround (De Angelis et al., 2017; Kapferer & Michaut-Denizeau, 2014; Torelli et al., 2012). Consumer demand is also a driving force, with a significant segment considering sustainability a key factor in their purchasing decisions (Kotler, 2011).

Thanks to information gathered from primary sources (mainly site visits, participation in seminars, and direct interviews) and secondary sources (mainly sustainability reports and information from company's social media), this chapter focuses on an exemplary case study that exemplifies the concept of retrovation within the apparel industry. This is the case of Rifò,[1] a business in the historical Prato textile district in Italy, where traditional *cenciaioli* artisans have revitalized their practices. These artisans are renowned for their expertise in selecting and repurposing textile scraps and waste, embodying the essence of retrovation by integrating traditional knowledge, innovation, and modern environmental and pro-social objectives.

Through this case study, this research highlights the potential of retrovation to drive circularity-driven sustainability in the apparel industry. It underscores the significance of leveraging traditional knowledge and practices to pave the way toward a more sustainable and circular apparel ecosystem. The revitalization of traditional practices toward innovation, or retrovation, offers a promising pathway for achieving sustainability in an industry often criticized for its environmental and social footprint.

2 Components of Circular Business Models in the Apparel Sector

Circular business models (CBMs) in the apparel industry encompass various components aimed at minimizing waste and maximizing resource efficiency throughout the product life cycle (Dragomir & Dumitru, 2022). CBMs are predicated on the 9R framework of circular economy that stands for the most to the last recommended action being Refuse, Rethink, Reduce, Reuse, Repair, Refurbish, Remanufacture, Repurpose, Recycle, and Recover (Potting et al., 2017). Knowing this, after "refuse" as the main attitude for CBMs, "rethink" is the second in order of importance. In the apparel industry, this means that designing clothes that last longer and are easier to repair are paramount to create a value chain closer to sustainability. The goal is to develop innovative business models that enhance consumer value while maximizing the long-term value of goods, components, and materials. This approach aims to

[1] Rifò is a Tuscan inflection of the verb *rifare* (meaning "to remake, to redo"), a name that takes up the tradition of textile recycling born in Prato and which has been handed down for over a century.

reduce the overall resource consumption, waste, emissions, and pollution (Konietzko et al., 2020).

As articulated by Geissdoerfer et al. (2018), CBMs can be regarded as a subset of sustainable business models (SBMs) due to their sustainability benefits, which stem from reduced resource inputs and lower waste and emissions. This assertion implies that circular supply chain management (CSCM), aimed at promoting sustainable development, should incorporate not only the typical features of a circular economy but also the distinctive characteristics of SBMs. These characteristics include adoption of a long-term perspective, proactive multi-stakeholder management, and creation of additional value through environmental and social sustainability considerations.

Thus, the circular economy represents a departure from the unsustainable, traditional linear fashion model ("take-make-waste"). The infeasibility and risks associated with the linear model are increasingly apparent, given its operational challenges, such as sourcing virgin raw materials, and strategic concerns, like the changing consumer preferences—almost 90% of Europeans believe clothes should be made to last longer (European Commission, 2022). As early as 2018, companies were advised to "self-disrupt their own identity and the sources of their old success" by adopting new business models aligned with sustainability concerns (McKinsey & Company and Business of Fashion, 2022). In this work, we are considering features of CBMs: design for longevity, circular production processes, and material innovation.

2.1 Design for Longevity

One fundamental component of circular business models in the apparel industry involves designing for durability and longevity, ensuring that garments withstand multiple uses and wash cycles (De Falco et al., 2019). Additionally, the utilization of sustainable materials, such as recycled cotton, reduces reliance on virgin resources and minimizes environmental impact (Chua, 2018). However, the recycling of polymeric fabrics is not so sustainable since it has microplastics as sub-products, also generated in the washing process of this clothes (Periyasamy & Tehrani-Bagha, 2022; De Falco et al., 2019; Napper & Thompson, 2016). Knowing this, garments must be designed to last long and for this the choice of the materials is paramount.

Schumann et al. (2021, p. 447) states that "the main problem in textile recycling remains the separation of the different fiber blends", which leads to the reflection that brands must start to design their clothes with mono-material fabrics, but not only this, because there are fabrics that are more sustainable than others based on their durability and composition. Thinking about a broader use of the term redesign, not only the conception of products must have new approaches but also the business models of companies that intends to be sustainable.

2.2 Circular Production Processes

According to the Ellen MacArthur Foundation (2021), a circular economy for fashion fosters the creation of superior products and services for consumers, presenting a significant growth opportunity that not only contributes to a resilient and thriving industry but also regenerates the environment. This regeneration is achieved by reducing waste and pollution, utilizing biodegradable resources, and employing renewable energy. Moreover, it enhances societal well-being (Fig. 1). Essentially, this model emphasizes the reuse of existing resources within the industry: extending the lifespan of clothing; facilitating the transfer of garments between users; and offering products and services that integrate, enhance, and replace garments in line with consumer needs (Ellen MacArthur Foundation, 2021).

However, rethinking the design of fashion supply chains is important to unlocking the potential of these new models. As highlighted by the Ellen MacArthur Foundation (2021), particularly for larger companies, circular business models are often implemented merely as "add-ons" to linear models, whereas they should entail a shift to an entirely new system (Ellen MacArthur Foundation, 2017). Close collaboration with upstream suppliers and retailers is crucial in devising new solutions that increase the longevity of clothing life cycles and retain both control and the additional value of products post-sale (Global Fashion Agenda and McKinsey & Company, 2020).

Nevertheless, textile recycling represents only 1% of the post-consumption treatment. Schumann et al. (2021, p. 184) argued that "[m]echanical recycling of old textiles remains tear fibres are obtained from the remaining old textiles, which are

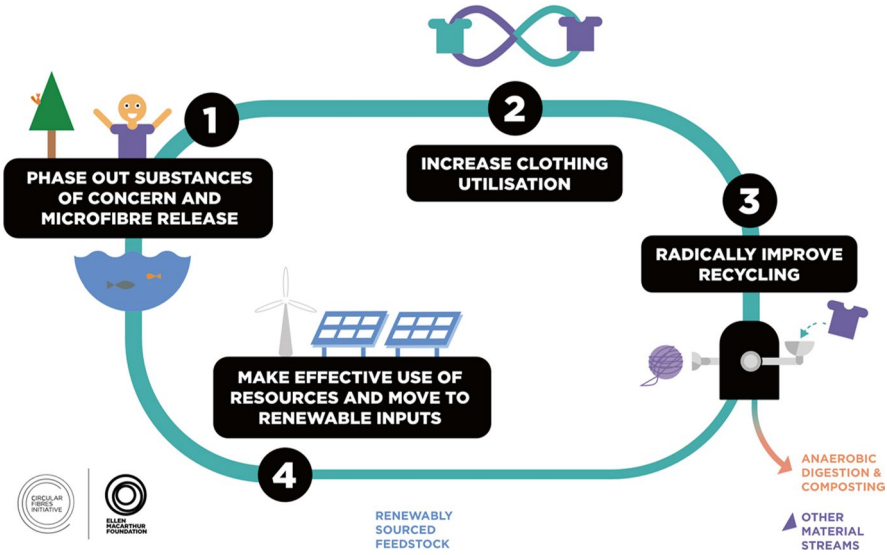

Fig. 1 Circular economy for fashion. (Source: Ellen MacArthur Foundation, 2017)

used almost exclusively in nonwovens for insulation, upholstery and automotive textiles. High-quality applications have so far been niche solutions". More engagement of producers and consumers must exist to enhance the volume of textile recycling and new uses to the recycled pieces must be discovered.

2.3 Material Innovation

In addressing the topic of material innovation for sustainable use in the apparel industry, it is essential to distinguish between natural and manufactured fibers. Natural fibers are derived from plant sources, such as cotton, linen, hemp, jute, and natural bamboo, or animal sources, including wool, silk, and cashmere. Conversely, manufactured fibers are synthesized from raw materials procured from diverse origins. These origins can be natural, as exemplified by fibers such as viscose, modal, and lyocell, or synthetic polymers, such as polyester, nylon, acrylic, polypropylene, and PVC. Contrary to conventional assumptions, natural materials do not inherently represent the superior or more environmentally sustainable option.

Paradoxically, polyester is occasionally marketed as a more sustainable alternative to cotton and wool. This assertion is due to the fact that polyester production necessitates significantly lower water usage compared to cotton—the global average water footprint for producing 1 kg of cotton is approximately 10,000 liters (Freitas et al., 2017), whereas the production of 1 kg of polyester reportedly requires negligible to no water (Fletcher, 2008). Additionally, polyester production does not involve the use of pesticides, further contributing to its perceived environmental benefits.

Nevertheless, the production of manufactured fibers is characterized by an extremely energy-intensive and polluting process. Additionally, the laundering of these materials has been found to be particularly detrimental to the environment, as they shed microplastic fibers that eventually accumulate in seawater (European Commission, 2022). Considering these aspects, numerous fashion brands have begun to introduce "greener" alternatives. These alternatives include organic cotton, which is not only beneficial for the environment but also features a strong social component linked to Fair Trade and ethical production. Furthermore, sustainable synthetics, such as Econyl®—produced through the recycling and purification of ocean and landfill waste have been developed. However, these sustainable options will necessitate significant technological innovations to achieve commercial viability on a large scale.

In this context, it is evident that different regulatory frameworks played a significant role for circularity and the specific case of the apparel sector. The European Green Deal (2019), more than other actions, has been a fundamental milestone. It is a package of policy initiatives, which aims to set the European Union on the path to a green transition, with the ultimate goal of reaching climate neutrality by 2050. It supports the transformation of the EU into a fair and prosperous society with a modern and competitive economy (European Commission, 2019). After this and

following the establishment of the Sustainable Development Goals in 2015, some forms of policies for circularity, for businesses, and specifically for the apparel sector have recently followed: important steps have been the Circular Economy Action Plan in 2020; the Ecodesign for Sustainable Products Regulation in 2022; the Sustainability Reporting Directive in 2023; the Green Claims Directive Proposal in 2023; and the Strategy for Sustainable and Circular Textiles in 2023.

3 Collection of Information for the Analysis

The result of two field visits in Prato by one of the author in 2024 is presented, along with two workshops organized by another author at the University of Padova, the first in November 2023, titled "Achieving Carbon Neutrality: Sustainable Business Models, Social Entrepreneurship, and B Corps", with the participation of Rifò's sustainability and innovation specialist, the second in December 2023, within the Sustainable Business Models course, featuring Rifò's founder as guest lecturer.

In addition, we also administered two direct interviews. The first one was conducted with the founder of Rifò in November 2022, when the company was participating in a start-up acceleration program in Florence (promoted by the innovation center Nana Bianca). The second interview was conducted with the firm's sustainability and innovation specialist in May 2024. The time lapse between the two interviews allowed, on the one hand, for the updating of information about the company and, on the other hand, for outlining its development patterns over time.

Finally, we elaborated on two more sources of information. On the one hand, information obtained in an interview in June 2024 with the manager of a textile history museum in Prato, the industrial district where Rifò is located. This interview was specifically important for gaining a broad perspective on the role of *cenciaioli* in the Prato district. On the other hand, we used data provided in the most recent Sustainability Report (2023) by Rifò with a detailed description of members of the company, used materials, obtained certifications, products, business commitment to sustainability and its impact, as well as projects for the future.

4 The Role of Retrovation for Circular Business Models: The Case of Rifò

Rifò is an emerging brand of clothes and accessories committed to producing high-quality products made with recycled textile fibers, all crafted in Italy, specifically in the industrial district of Prato, one of the most iconic mature industrial districts specialized in the textile sector in Europe (Bellandi and Storai, 2022). This company was mentioned by the *Financial Times* among the 1000 fastest-growing companies in Europe, ranking 253rd (Financial Times, 2024). This recognition brings

great satisfaction at Rifò but also awareness: in fact, they suggest that economic growth should not be the primary goal. Therefore, in their company's supplementary contract, Rifò has included not only revenue targets but also sustainability and textile recycling goals. As they suggested in Rifò (2024), "[g]rowth and change must go hand in hand".

Utilizing a mechanical and artisanal process inspired by what was developed over a century ago in the textile district of Prato, Rifò transforms fabric leftovers and old clothes into new garments that retain the same quality as the original products. This method has a strong sustainable sense as it enables the recycling of textile waste while significantly reducing the use of water, dyes, and chemicals in production. In a market where it is easy to find affordable and attractive clothing, Rifò believes in doing better by offering high-quality, fashionable attire that also cares for the planet. Their mission is to create a line of stylish and high-quality garments made entirely in Prato from 100% recycled and recyclable fibers.

Established in 2018, the company presently employs 25 people, with 19 being permanent employees. Remarkably, 72% of its workforce is on permanent contracts, and women make up 72% of the staff, with 55% in managerial roles. This gender-inclusive approach highlights Rifò's commitment to a diverse and stable work environment. The company's turnover now exceeds €3 million annually, with 60% of its products sold within Italy and significant exports to Germany, Austria, Switzerland, and France.

Rifò's business model is inherently circular, diverging significantly from the traditional linear models of production and consumption. Key innovations focus on several areas: process innovation through implementing new technologies while maintaining in-house design and outsourcing production; business model innovation by adopting a circular model that revives traditional practices of recycling and reusing materials; and service innovation through developing new customer services that align with sustainable practices. A brief idea of their business model is presented in Fig. 2.

Central to Rifò's strategy is reviving and modernizing heritage techniques and craftsmanship to align with contemporary sustainability objectives. By seamlessly blending tradition with modernity, Rifò integrates time-honored skills like those of the *cenciaiolo* (ragman), shown in Fig. 3, spinner, and weaver into their processes. The *cenciaioli* traditionally used their experience, touch, and manual skills, as well as their sight, to divide bundles of fabrics and fibers into homogeneous groups. In the case under study, they emphasize the reintroduction of traditional mending skills, aiming to create a socially responsible knitting factory. This blend of past and present fosters sustainable innovation, reinforcing the intrinsic link between heritage, sustainability, and innovation.

The retrovated activity of the *cenciaioli* is a specific case of Rifò. In fact, as stated by the manager of a textile history museum in Prato, the activity of the *cenciaioli* has now been lost in Prato and replaced by machinery and automation.

Rifò's commitment to sustainable production is evident across all stages of their operations: products are designed using recycled fibers, with a minimal addition of virgin or organic components to enhance durability. They avoid prints, as garments

Fig. 2 Rifòs's circular business model. (Source: Rifò's website, 2024)

Fig. 3 Rifò's retrovated *cenciaiolo* (ragman). (Source: Rifò's website, 2024)

with prints are less recyclable, and use natural dyes. Additionally, the company minimizes the use of accessories to ensure higher recyclability of their products. The production process involves selecting local artisans who meet Rifò's stringent standards, allowing better quality control and supporting local craftsmanship.

Rifò's supply chain is strictly local, with regular visits to collaborators to ensure adherence to their code of conduct. Bureau veritas conducts semi-announced audits

to verify compliance with the International Labour Organization's principles in the field of sustainability. Moreover, the company holds a B-Corp certification since November 2020, underscoring its commitment to high social and environmental performance standards.

In fact, from the interviews, it emerged that Rifò generates 100% of its revenue from goods produced using sustainable methods. Specifically, the company sources raw materials carefully, with sustainability being a key criterion. For instance, they use Global Recycled Standard (GRS) certified materials, which uphold both environmental and social standards. Moreover, it employs innovative natural dyeing techniques, using dyes from peach nuts, walnuts, and chlorophyll pigment, as well as mineral dyes from volcanic rocks. These techniques produce unique, albeit limited, color ranges suitable for small-scale production, adding a distinctive touch to each product. The company also promotes recycling and upcycling. They purchase yarns with recycled fibers and run take-back programs for old jeans and sweaters from customers. Additionally, leftover yarns are used in their *zero waste collection*, creating colorful accessories like hats and gloves.

To engage customers, Rifò uses social media and newsletters to promote and educate about sustainable consumption. In fact, they suggest that their circular service model can further educate and involve customers in sustainable practices. The company also collaborates with other organizations and brands to enhance its sustainable practices, with notable partnerships including Fideuram, Levi's, Bettaknit, and Kering. One outstanding initiative to collaborate with the active participation of consumer in the recycling process is spreading jeans drop-off collection points, as shown in Fig. 4, in purpose-aligned partner stores as NaturaSì and Muji.

Despite its successes, Rifò faces significant challenges, primarily related to scaling up while maintaining sustainability standards. As sustainable fashion grows, maintaining these standards on a larger scale becomes increasingly difficult. Looking ahead, Rifò envisions a significant role for the EU's strategy for

Fig. 4 Rifò's spreading jeans drop-off collection points. (Source: Rifò's website, 2024)

sustainable circular textiles in promoting sustainability. The future of sustainable fashion, according to Rifò, hinges on a shift in consumer habits and business models toward sustainability.

5 Discussion

As introduced in the previous section, we suggest that Rifò exemplifies how traditional practices can be revitalized to foster circularity and sustainable innovation in the modern apparel industry. Through a combination of process, product, and service innovations, the company meets the current demand for sustainable fashion and could set a benchmark for future practices in the industry. By embracing retrovation, Rifò demonstrates the powerful synergy between circularity, innovation, and heritage, showcasing how revisiting and modernizing the interpretation of traditional methods can drive sustainable transformations in the apparel sector.

The transformation toward circularity in the apparel industry, as exemplified by Rifò, demonstrates a shift from traditional sustainable adjustments to innovative business models that integrate heritage techniques with modern circular practices. This shift is significant because it moves beyond mere resource efficiency to encompass a holistic approach that includes manual textile elaboration and natural dyeing processes, which have been largely overlooked in contemporary mass production.

The integration of retrovation within Rifò's business model showcases how traditional practices can be revitalized and adapted to meet modern sustainability goals. This approach not only preserves cultural heritage but also enhances the value proposition of sustainable fashion by creating unique, high-quality products that appeal to environmentally conscious consumers. The emphasis on local artisanship and traditional skills, such as those of the *cenciaioli*, reinforces the importance of community and local economies in achieving sustainability.

Rifò's commitment to sustainability is evidenced in their selection of natural materials such as wool and cashmere, avoiding the generation of microplastics by petroleum-derivative fabrics like polyester and acrylic. Another strategic practice is minimizing the use of accessories to enhance recyclability. With this, Rifò sets a high standard for sustainable fashion. Their supply chain, which is strictly local and adheres to rigorous sustainability criteria, further underscores the feasibility and benefits of a circular economy in the apparel industry.

The empirical evidence from Rifò's case study highlights the practical implications of retrovation in originally driving circularity. The company's ability to blend traditional techniques with innovative business strategies may serve as a benchmark for other firms in the industry. This approach not only addresses the environmental impacts of fashion production but also contributes to social sustainability by supporting local craftsmanship and providing stable employment opportunities.

However, the scalability of such models remains a challenge. As sustainable fashion gains traction, maintaining high standards of sustainability while expanding operations is increasingly complex. Rifò's proactive adaptation of their business

model to incorporate sustainability goals alongside economic targets is a crucial strategy in navigating these challenges.

The broader implications of this study suggest that the successful integration of retrovation and circularity can serve as a catalyst for industry-wide change. By demonstrating the viability of sustainable practices rooted in traditional knowledge, Rifò can provide a roadmap for other companies to follow. This case study reinforces the potential of heritage practices to drive innovation and sustainability, offering a promising pathway toward a more circular apparel industry.

Specifically, the discussion on the potential of retrovation within circular business models for the apparel sector reveals several key insights. First, the reintegration of traditional practices not only promotes environmental sustainability but also strengthens cultural heritage and community engagement. The *cenciaioli* artisans in Prato exemplify how historical craftsmanship can be harnessed to meet contemporary sustainability goals.

Moreover, Rifò's approach underscores the importance of local supply chains in achieving sustainability. By sourcing materials locally and maintaining close relationships with artisans, the company reduces its carbon footprint and supports local economies. This localized approach contrasts sharply with the globalized nature of the traditional fashion industry, which often prioritizes cost reduction over environmental and social impacts.

Another crucial aspect of Rifò's strategy is its focus on consumer education and engagement. By involving customers in recycling programs and emphasizing the story behind each product, Rifò fosters a deeper connection between consumers and their purchases. This engagement not only builds brand loyalty but also promotes a broader cultural shift toward more sustainable consumption patterns.

The challenges associated with scaling up such sustainable models cannot be overlooked. As the demand for sustainable fashion grows, companies like Rifò must find ways to expand their operations without compromising their sustainability principles. This requires continuous innovation in production processes, supply chain management, and business strategies. It also necessitates collaboration with other stakeholders, including policymakers, industry peers, and consumers, to create an enabling environment for sustainable growth.

6 Conclusion

In the time of another great challenge driven by hyper-fast fashion and large amounts of textile waste, the apparel industry is looking for new models of circular textile elaboration and production (or even slow fashion). In front of the linear models of extraction, processing, design, manufacturing, retail, use, end of life, and disposal, new phases as introduction of renewable or bio-based inputs, recycling, remanufacturing, reuse, repair, industrial symbiosis, and sharing could be included (Mhatre et al., 2021).

In this context, we introduced the case of retrovation and we analyzed the company Rifò, where this model has been implemented. The case of Rifò demonstrates the transformative potential of combining retrovation with circular economy principles in the apparel industry. By leveraging traditional knowledge and practices, companies can create innovative and sustainable business models that address both environmental and social challenges. More broadly, this company exemplifies the potential of combining traditional practices with modern sustainability initiatives to create a circular economy in the apparel industry.

Rifò's experience provides valuable lessons for the industry, highlighting the need for a holistic approach that integrates sustainability at every level of the value chain. This approach not only addresses environmental and social challenges but also enhances consumer value and supports local economies.

This model highlights the importance of localizing supply chains to minimize carbon footprints and promote local economic strategies. For example, by sourcing materials locally and utilizing traditional artisanal skills, Rifò not only reduces transportation emissions but also supports the livelihoods of local artisans. This approach fosters a symbiotic relationship between sustainability and local culture, enriching the value proposition of sustainable fashion. Furthermore, Rifò's dedication to natural materials and minimizing the use of accessories ensures that their products are not only environmentally friendly but also easier to recycle. This careful consideration of product life cycle from design to disposal exemplifies how circular economy principles can be integrated into every stage of the production process. Such practices set a high standard for the industry and offer a replicable model for other companies seeking to enhance their sustainability credentials.

The broader implications of Rifò's success suggest that the integration of traditional practices and current (and future) sustainability initiatives can serve as a powerful catalyst for industry-wide change. By demonstrating that heritage techniques can interact with innovative business strategies, Rifò can provide a roadmap for other companies to follow.

However, the scalability of such models remains a significant challenge. As sustainable fashion gains traction, maintaining high standards of sustainability while expanding operations becomes increasingly complex. Rifò's ability until now to navigate these challenges through continuous innovation and adaptation can serve as a crucial strategy for other firms. It is relevant that these companies engage in ongoing research and development to refine their processes and maintain their commitment to sustainability.

Consumer education and engagement are also critical components of this transition. Efforts to involve customers in recycling programs and emphasize the story behind each product help foster a deeper connection between consumers and their purchases. This engagement not only builds brand loyalty but also promotes a broader cultural shift toward more sustainable consumption patterns. Educated consumers are more likely to support sustainable brands, creating a virtuous cycle that benefits both the industry and the environment.

The pursuit of sustainability in the apparel industry is not an isolated endeavor. All stakeholders in the production and consumption chain must share responsibility.

Despite the current limited presence of very stringent directives and regulations from governments and international organizations for mass production, and the low level of motivation and awareness among the majority of consumers to curb consumerism, some pioneering companies committed to sustainability must proactively adapt their business models. These adaptations aim to reduce negative externalities on the environment and society.

Relevant implications emerge. Particularly, policymakers must create an enabling environment through supportive regulations and incentives, while industry peers should collaborate to share best practices and drive collective progress. Consumers, too, play a pivotal role by making informed purchasing decisions and supporting brands that prioritize sustainability.

This study contributes to the growing body of evidence supporting the viability and benefits of circular business models, offering a roadmap for others in the industry to follow in the pursuit of a more sustainable future. The collective effort of all stakeholders, businesses, consumers, and policymakers will be crucial in driving this transformation and ensuring a sustainable and prosperous future for the apparel industry. From the theoretical point of view, this work offers one of the rare empirical studies on the concept of retrovation, which has recently emerged as an important pattern in the more established field of responsible innovation for addressing grand societal challenges (Sedita, 2024). Specifically, it provides a specific case study of retrovation in the realm of circularity and sustainability. Moreover, based on an innovative interpretation of practices typical of certain local heritages, it also suggests a preliminary link between the concept of retrovation and studies of local economic development where productive traditions and territorial/sectorial identities play a fundamental role.

There are limitations in this study. As previously mentioned, the case of Rifò suggests tools for applying retrovation by mixing traditional techniques and practices with innovation from a sustainability perspective. However, it cannot be claimed that other companies or the entire sector operate in the same manner. Therefore, the case under analysis might serve as an example for other companies in the apparel sector for the future. This also suggests interesting avenues for future research, which include expanding the sample of interviewed companies—also in other contexts—and empirically validate the assertions made in this chapter, thereby enhancing the practical applicability of our theoretical framework.

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